

Yukhta Seelam

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SUMMARY

Graduate student in Robotics and Autonomous Systems at Boston University with hands-on experience in robotic hardware design, automation, and embedded systems. Skilled in end-to-end development of electromechanical systems including CAD design, control integration, and prototyping for real-world applications.

ACADEMIC PROFILE

Boston University **3.8/4.0**
Masters in Robotics and Autonomous Systems

Expected Jan 2026

Indian Institute of Technology Kanpur
B. Tech in Electrical Engineering

May 2024

EXPERIENCE & PROJECTS

Dexter Research Rover — Equipe de Robotique Autonome (ERA-IITK)

Dec 2023 – May 2024

- Designed and assembled a modular mobile robot with power management, motor control, and embedded microcontroller systems.
- Created 3D CAD models in Fusion360 and implemented power/BMS circuits using Eagle and Arduino.
- Integrated a 3-DoF autonomous robotic arm with servo-electric gripper for pick-and-place operations.

Samsung Research & Development – India

May 2023 – Jul 2023

- Supported cross-functional hardware and AI development for next-gen robotic vacuum cleaners.
- Validated sensor integration and developed optimized detection models to enhance perception and motion planning accuracy.

Development of AI Combat Robots — IIT Kanpur / ICRA 2022 RoboMaster Challenge

Mar 2022 – Jan 2023

- Engineered two AI-enabled combat robots with NVIDIA-embedded GPUs, achieving high precision in motion control and targeting.
- Built mechanical chassis and gimbal systems, integrating LiDAR and camera-based localization using ArUco markers.
- Led hardware debugging and real-time system calibration, enabling 90% accuracy during dynamic firing tests.

Course Designer & Mentor — IIT Mandi

Mar 2022 – Aug 2022

- Designed a robotics practicum course combining mechanical assembly, sensor integration, and ROS-based control.
- Supervised students in designing 3D-printed components and assembling autonomous robots with microcontrollers.

Obstacle Avoidance and Path Planning for Rover — IIT Kanpur

Dec 2021 – Feb 2022

- Designed mechanical and sensing systems for an outdoor rover navigating uneven terrains.
- Implemented real-time motion control using ROS and integrated sensor fusion modules for environment mapping.

AWARDS AND PUBLICATIONS

- 3rd Place – IEEE ICRA DJI RoboMaster AI Challenge 2022 (only Indian team internationally)
- Paper: “Towards Autonomous Shooting Rover via Situation-Aware Visual Perception and Dynamic Action Execution”, IEEE/SICE SII 2024 (Vietnam)
- Poster: “Design of AI Ground Robots for Intelligent Combat in a Dynamic Arena”, ICRA 2022

LEADERSHIP POSITIONS

- Team Head – Robotics Club - IITK | President – Toastmasters IIT Kanpur | Captain – IITK Basketball Team | Student Ambassador – BU Robotics

SKILLS & COURSEWORK

- Mechanical & Design:** Fusion 360, SolidWorks, Abaqus, AutoCAD, Eagle, 3D Printing, CAD/CAM
- Robotics & Embedded:** ROS, Gazebo, Arduino, C++, MATLAB, Control Systems, Sensors & Actuators
- Programming & Tools:** Python, C++, Bash, SQL, Git
- Manufacturing Processes:** Prototyping, Testing, Mechanical Assembly, PCB Design